Portfolio Optimization and Client Segmentation

Group 7

ISBA 2405

Due Date: 6/6/2025

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1. Project Overview

In this project, we address the challenge of personalized investment strategies through portfolio optimization and customer segmentation. In today's financial environment, clients' risk tolerance, liquidity requirements and investment horizons vary widely, so investment options are also very broad and complex. Our goal is to use cluster analysis, optimization modeling and scenario testing to design customized investment portfolios that meet these different characteristics, thereby helping to provide more suitable options for different customer groups and different market investments.

2. Methodology Summary

Our approach follows a structured, multi-stage analytical process, then combines powerful data processing with advanced modeling techniques. First, we perform rigorous data cleaning and transformation, followed by exploratory data analysis (EDA) to understand asset behavior and customer profiles. We then apply K-means clustering(Elbow Method) to both datasets to divide into meaningful groupings. Finally, we develop and implement multiple optimization models—including baseline, scenario-based, customer-specific, and multi-objective models—to construct personalized and risk-consistent portfolios.

3. Analysis

3.1 Exploratory Data Analysis (EDA)

To ensure the accuracy of the analysis, we performed the following four steps to lay the foundation for modelling analysis:

- Data Quality Check: First, we validated the asset and customer datasets by identifying and removing missing values and duplicate records, which ensured that the asset_returns dataset was complete, numerically consistent, and free of outliers to avoid impact.
- Data Construction: We converted the separate year and month fields to a standardized date format for accurate time series alignment of return and performance analysis. This conversion enables us to effectively capture time trends and cumulative return dynamics.
- Classification Standardization: We cleaned and standardized key classification variables such as Asset_Class and ESG_Preference to eliminate format inconsistencies, fix case sensitivity issues, and reduce space errors. This step is critical to maintaining consistent grouping, filtering, and modeling.
- Visualization and Statistical Analysis: We visualized the core asset indicators and customer characteristics to provide information for following clustering and optimization logic:

Asset-Level EDA:

1. Monthly Average Return

From the monthly returns of each asset class, the chart shows that technology stocks, Apple (S001) and Microsoft (S002) have the highest average returns, especially Apple, with an average monthly return of more than 3.5%. On the contrary, for example, the returns of US 10-year Treasury bonds (B001) and corporate bonds (B002) are quite stable, but not high, with an average monthly return of less than 0.5%.

These two types of asset classes show that growth-oriented portfolios can obtain capital appreciation by holding technology stocks, while conservative portfolios can obtain stable returns by holding fixed-income assets.

2. Volatility Insight – Return Distribution Patterns

When we examine the monthly return distribution. Some assets such as Apple, Microsoft, and emerging markets have long and highly skewed return distributions, which indicate higher volatility and greater upside or downside potential. Meanwhile, bonds and REITs have narrower and symmetrical distributions, mostly centered around zero, this indicates the lower volatility and more stable performance.

This analysis was later incorporated into the risk segmentation model to find aggressive clients holding high variance assets and conservative clients holding low variance assets.

3. Cumulative Return Over Time

For the cumulative returns, which shows that technology stocks did better than other sectors lately. Companies like Apple and Microsoft had strong growth, showing they can really pick up speed during this time. On the other hand, U.S. Treasuries and corporate bonds didn't perform as well, with slower increases in value.

This helps explain why many long-term clients are being encouraged to put more of their investments into stocks.

4. Correlation Heatmap – Diversification Potential

In addition, we also constructed a correlation matrix to analyze the relationship between asset returns. The chart shows that there is a significant correlation between technology stocks, like Apple and Microsoft, their correlation coefficient is more than 0.7, which shows that these assets have the characteristics of growth-oriented aggregation. In contrast, bonds and gold have low or negative correlations with stocks.

This illustrates the value of bonds and gold in diversification and supports their inclusion in risk hedging portfolios, thereby achieving the purpose of preventing risks and reducing volatility.

Client-Level EDA:

1. Risk score distribution

The client risk score ranges from 1 to 10, and the index can reflect the wide range of risk tolerance of clients. We observe in the chart that it is basically concentrated around 2-3 points and 5-8 points, which shows that investors are both conservative and moderately aggressive. The variety of risk preferences means the importance of developing personal portfolio strategies to support different levels of risks.

2. Age distribution

The chart shows that most clients are between 30 and 60 years old, which is also the actual working-age population structure in today's market. This has a direct impact on investment planning: younger clients are more likely to prefer growth-oriented strategies to earn more money, while older clients usually seek income stability and capital preservation as they approach retirement.

3. Net worth distribution

Most clients have a net worth between $400,000 and $1 million, with only a very small number of high net worth clients having a net worth of more than $1.2 million. This can help move from simplified, low-cost solutions for mid-tier clients to more diversified, complex portfolios for wealthy individuals.

4. Liquidity needs

Liquidity preferences are concentrated at low to medium levels, which indicates that most clients don't need short-term cash withdrawals. This allows them to focus more on long-term investments and compounding strategies, making growth and medium-liquidity assets suitable for a considerable portion of clients.

3.2 Clustering Analysis

To support personalized portfolio construction, we applied K-means clustering separately to both the asset characteristics and the client profile. The optimal number of clusters was determined using the **Elbow Method**.

Asset Clustering (k = 3): Using asset-level return and volatility	Client Clustering (k = 3): Based on client profile
• **Cluster 0 – Low-risk, stable assets:** This group includes assets such as US	• **Segment 0 – Moderate Profile:** Clients in this category have a moderate risk

Treasury Bonds and corporate bonds which exhibit the low volatility and average returns which is ideal for conservative, risk-averse portfolios.	tolerance, a longer investment horizon (approximately 14.5 years), and low liquidity needs. They have an average net worth of nearly $1 million, are financially stable, and are suited to a diversified, balanced growth portfolio.
• **Cluster 1 – Moderate-risk, growth-oriented assets:** These assets like Johnson & Johnson, Procter & Gamble, and REITs, which offer a moderate return and volatility which is suitable for balanced investment strategies.	• **Segment 1 – Aggressive Profile:** Clients in this category have a higher risk profile, a longer investment horizon (around 28 years), and no short-term liquidity needs. However, because their net worth is typically lower, they are best suited to high-growth, high-volatility, and long-term return portfolios.
• **Cluster 2 – High-risk, high-return assets:** This area is dominated by technical stocks like Apple Inc. and Microsoft Corp.They deliver the highest returns and volatility, which can satisfy the needs of aggressive or long-horizon investors.	• **Segment 2 – Conservative Profile:** This category of clients has a lower risk tolerance, a shorter investment horizon (around 5 years), but higher liquidity requirements. Based on their moderate net worth, they require an investment portfolio that focuses more on capital protection and short-term liquidity.

Optimization Models

4.1 Base Model – Mean-Variance Optimization

We implemented the classical Markowitz mean-variance optimization model to serve as the foundation for our portfolio construction framework.

Model Methodology:

- **Objective Function:** Aiming to minimize portfolio variance, while meeting a target expected return defined as the average return across all assets.
- **Decision Variables:** Asset weights $w_i \in [0, 1]$ where $\Sigma w_i = 1$
- **Key Constraints:** The covariance matrix and historical mean returns of assets were used to compute the risk and return profile.

Optimization Results	Asset Allocations
Expected Return: 1.32%	US Treasury 10-Year: 48.8%
	Microsoft Corp: 25.0%
Portfolio Variance: 0.0533%	Johnson & Johnson: 22.5%
	Apple Inc: 2.9%
	JPMorgan Chase: 0.8%

Insight:
The model allocated nearly half of the portfolio to the U.S. Treasury bonds, prioritizing downside protection and portfolio stability. Microsoft and Johnson & Johnson were selected for their reliable returns and moderate volatility, while Apple and JPMorgan were included in smaller proportions for exposure to growth and financial sectors. This portfolio composition reflects a conservative stance with modest return expectations and minimal risk.

4.2 Scenario Analysis

To evaluate the robustness of portfolio performance under different macroeconomic environments, we conducted scenario-based optimization using six representative economic conditions. In each case, expected returns for assets were adjusted according to assumed market reactions, and portfolio weights were re-optimized.

Scenario	Expected Returns	Asset Allocation
Base Case	16.70%	S001(Apple Inc.)
High Growth	17.37%	S001
StagFlation	14.86%	S001
Market Correction	13.19%	S001
Severe Recession	10.52%	S001
Deflation	14.36%	S001

Observation:
Across all economic scenarios, the model mainly allocated the entire portfolio to Apple Inc. (S001), which indicates its superior performance when optimized solely for expected return.

The High Growth scenario generated the highest return of 17.37%, which is driven by favorable assumptions regarding technological sector expansion.
In contrast, the Severe Recession scenario produced the lowest return with 10.52%, which emphasizes the sensitivity of equity assets to macroeconomic depression
Then for the Intermediate scenarios, including Stagflation and Deflation, which generate moderate returns ranging from 14.0% to 15.2%, they reflect mixed economic signals and sector-specific performance variations.

Insight:
While Apple Inc. dominates across all market assumptions, this lack of diversification introduces risk. The results highlight the importance of integrating additional constraints—such as asset caps or risk balancing—to prevent over-concentration, even when a single asset appears optimal under purely return-based criteria.

4.3 Diversification and Risk-Based Models

To explore risk segmentation and its impact on portfolio design, we created three models based on the K-means clustering results of the assets:

- **Cluster 0 (Low-Risk Assets):** Primarily composed of US Treasury Bonds and Corporate Bonds
- **Cluster 1 (Moderate-Risk Assets):** Including REITs, Johnson & Johnson, and Procter & Gamble
- **Cluster 2 (High-Risk Assets):** Represented mainly by Apple Inc. and Microsoft Corp.

Each model was restricted to use assets only from one cluster, simulating how clients with different risk profiles would invest:

a. High-Risk Portfolio (Cluster 2 only):

Optimization Behavior	With no constraints on diversification within the clusters, the model selected 100% allocation to S001(Apple Inc.), which had the highest expected return
Optimization Outcome:	**Expected Return:** 16.7%
	Portfolio Variance: Not explicitly reported, but implied to be high due to single-asset concentration

Insight:
While mathematically optimal in terms of expected return, this solution lacks diversification and poses considerable risk. It reflects the typical behavior of return-driven optimization when no risk controls or caps are imposed.

b. Moderate- and Low-Risk Portfolios:

Although the report screenshot focuses on Cluster 2, similar behavior was observed in constrained clusters as well. In several optimization runs, the solver defaulted to selecting a single top-performing asset within each cluster, particularly in unconstrained formulations. This further reinforces the insight that without explicit diversification constraints, optimization results often concentrate excessively on a few dominant assets.

Using the asset cluster classifications derived from K-means, we developed three distinct portfolios:

Portfolio Type	Asset Composition	Key Characteristics
Low-Risk Portfolio(Cluster 0)	Bonds and Gold	Optimized stability and capital preservation, also low volatility and returns
Moderate-Risk Portfolio(Cluster 1)	REITs and Blue-Chip consumer staples	Balanced risk and returns
High-Risk Portfolio(Cluster 2)	Growth Assets(Apple, Microsoft)	Maximized expected return with higher risk

These models can help match asset strategies to risk-segmented client groups.

4.4 Client-Specific Models

To demonstrate the benefits of personalization, we developed dedicated portfolio models for 4 groups of representative clients, each with unique financial characteristics, constraints, and goals. These models used client-specific information including risk tolerance, liquidity needs, and investment horizons, and applied them as constraints in the optimization.

To address diverse client needs, we designed 4 customized portfolio optimization models, each tailored to a distinct investor profile:

- Foundation Client:$\max(x)\ \lambda \cdot (i\sum \mu i xi) - (1-\lambda) \cdot (xT\Sigma x)$

- C1001 (Near-Retirement): Modeled with tight liquidity, dividend, and bond inclusion constraints, this client received a conservative, income-generating portfolio with balanced exposure to REITs, blue-chip stocks, and fixed income.
- C1002 (Young Professional): Optimized using a flexible MILP model focused on maximizing return with minimal restrictions, resulting in a concentrated, high-growth allocation to Apple Inc., reflecting the client's high risk tolerance and long investment horizon.
- C1003 (Balanced Strategy): Built with a structured LP model integrating lot size constraints, sector diversification, bond floors, and credit quality minimums. This client received a diversified and disciplined allocation balancing safety and growth across sectors.

Together, these models demonstrate how optimization can flexibly adapt to investor-specific constraints, yielding personalized portfolios aligned with each client's goals and tolerances.

a. Foundation Client (Baseline Advisory Case)

To establish a foundational investment profile applicable to general-purpose advisory needs, we developed a flexible yet disciplined optimization model targeting diversified growth.

Modeling Methodology:

- **Objective:** Maximize total expected return by using $430,000 budget
- **Decision Variables:**
  - x[i]: proportion of total capital invested in asset *i*
  - y[i]: binary variable indicating whether asset *i* is selected
  - z_alt[i]: integer lot counter for alternative investments
  - z_class[c]: binary indicator of whether asset class *c* is used
- **Key Constraints:**
  - Budget Constraint: Full allocation of the $430,000 investment budget.
  - Position Cap: Maximum of 10 unique assets selected.
  - Minimum Investment Threshold: Any selected asset must receive at least 1% of the portfolio.
  - Alternative Assets: Investments in alternative assets must be made in increments of $20,000.
  - Technology Exposure: If a technology asset is selected, it must comprise at least 8% of the portfolio.
  - Asset Class Diversity:At least 3 different asset classes must be represented.

Optimization Results	Asset Allocation

Expected Return: $71,289.70(16.57%)	Apple Inc.(S001): $421,400(98%)
	Johnson & Johnson(S003): $4,300(1%)
	JPMorgan Chase(S004): $4,300(1%)

Insight:
Despite the imposed diversification rules, Apple Inc. remained the dominant investment, capturing over 98% of the total budget due to its superior return profile. Minimal allocations to Johnson & Johnson and JPMorgan Chase satisfied class diversity and tech exposure requirements. This result underscores how a rules-based approach can guide portfolio construction without fully constraining return potential, making this foundation model a useful baseline for general clients.

b. Client C1001 – Near-Retirement Investor

Modeling Methodology:

- **Objective:** Maximize dividend income by using $850,000 budget
- **Key Constraints:**
  - Bond Inclusion: Portfolio must include at least one bond asset.
  - Bond Investment Granularity: Investments in bonds must be made in increments of $10,000.
  - Assets Minimum: If selected, assets must receive at least $15,000 allocation.
  - Equity Asset Limit: A maximum of five equity assets may be included in the portfolio.
  - Per-Asset Investment Range: Each selected asset must receive no less than $1,000, and no more than 70% of the total portfolio value.
  - Dividend Income Floor: The portfolio must generate at least $3,000 in annual dividend income.

Optimization Result	**Asset Allocation**
Expected Return: $127,720(15.03%)	Apple Inc.(S001): $340,000(40%)
	Microsoft Corp(S004): $340,000(40%)
	Corporate Bond Fund(B002): $10,000(1.18%)

Insights:
Despite the client's conservative nature, the model balanced risk and yield by combining

high-dividend REITs and blue-chip tech stocks with a small allocation to bonds. Dividend income was also optimized for liquidity needs.

c. Client C1002 – Young Professional

The portfolio optimization for Client C1002 was implemented using a Mixed-Integer Linear Programming (MILP) model via the Gurobi solver, with the objective of maximizing expected return under a set of client-specific investment constraints.

Modeling Methodology

- **Objective Function:**
  - Objective = sum over i of (Expected_Return[i] × x[i] × Total_Budget)
  - Where:
    - x[i] is the fraction of total budget invested in asset i
    - Expected_Return[i] is the average annual return of asset i
    - Total_Budget = $24,000
- **Decision Variables:**
  - x[i]: Continuous variable (between 0 and 1), representing the fraction of the total budget invested in asset i
  - y[i]: Binary variable, equals 1 if asset i is included in the portfolio, 0 otherwise
  - z_intl: Binary variable, equals 1 if any international asset is used
  - intl_total: Total fraction of investment allocated to international assets
- **Key Constraints:**
  - Budget Constraint: Sum of all (x[i] × Total_Budget) = 24,000
  - Minimum Investment per Asset: If asset i is selected (y[i] = 1), then x[i] × Total_Budget ≥ 5,000. If not, selected (y[i] = 0), then x[i] = 0
  - Enforced by: x[i] × Total_Budget ≥ 5,000 × y[i] , x[i] ≤ y[i]
  - Diversification Limit (Max 8 Assets):Sum of y[i] across all assets ≤ 8
  - International Asset Exposure:
    Let intl_total = sum of x[i] over all international assets.If any international asset is selected (z_intl = 1), then: 0.15 ≤ intl_total ≤ 0.25
  - If no international assets are selected (z_intl = 0), then: intl_total = 0, This is enforced by:intl_total ≥ 0.15 × z_intl ,intl_total ≤ 0.25 × z_intl. For each international asset i: z_intl ≥ y[i]

Optimization Result	**Asset Allocation**
Expected Return:$4,008 (16.70%)	Apple Inc.(S001): $24,000(100%)

Insights:
The model allocated the entire portfolio to Apple Inc., maximizing expected return due to the client's aggressive risk profile. While the return is high, this portfolio carries substantial volatility and is highly concentrated, which may require diversification caps in practice.

d. Client C1003 – Balanced Strategy

Modeling Methodology:

- **Objective:** Maximize return by using $24,000 budget
- **Key Constraints:**
  - Investment must be either $0 or ≥ $5,000
  - Must be either 0% or at least 15%
  - At most 8 assets allowed
  - $2,000 monthly contributions over a year

Optimization Result	**Asset Allocation**
Expected Return:$108,900 (10.89%)	Apple Inc.(S001): $24,000(100%)
	Johnson & Johnson(S003):$300,000 (30.0%)
	Corporate Bond Fund(B002):$250,000 (25.0%)
	Developed Markets ETF(I002):$50,000 (5.0%)

Insights:
This client received a well-diversified portfolio with exposure across sectors and risk classes. The mix of equities, fixed income, and international assets reflects a moderate growth strategy while ensuring downside protection.

4.5 Efficient Frontier – Multi-Objective Optimization

To explore the trade-off between return maximization and risk minimization, we implemented a multi-objective portfolio optimization model using a convex combination of expected return and portfolio variance.

The goal is to identify portfolios that optimally balance two competing objectives: maximizing expected return and minimizing portfolio risk. This balance is controlled through a tunable parameter **λ (lambda_risk)**:

- $\lambda = 0 \rightarrow$ focus entirely on return
- $\lambda = 1 \rightarrow$ focus entirely on minimizing variance
- $0 < \lambda < 1 \rightarrow$ balance between the two objectives

Mathematical Methodology:
Model Objective: This model balances expected return and portfolio variance using a risk-aversion parameter, lambda (λ), where λ ranges from 0 to 1. The mathematical objective function is: Objective = $\lambda \times (w^t \times \Sigma \times w) - (1 - \lambda) \times (w^t \times \mu)$

Where:

w is the vector of asset weights
Σ is the covariance matrix of asset returns
μ is the vector of expected returns
λ controls the trade-off between variance and return

The first term ($w^t \times \Sigma \times w$) represents the portfolio's total variance (risk), while the second term ($w^t \times \mu$) represents the expected return. This is a quadratic optimization problem solved using Gurobi.

Constraints:

1. **Fully Invested Portfolio: Sum of all weights $w_i = 1$**
2. **No Short-Selling : Each weight $w_i \geq 0$**.

Optimization Result(λ = 0.5)	**Asset Allocation**
Expected Return: 3.52%	Apple Inc.(S001): 91.32%
Portfolio Variance: ~0.78%	Microsoft Corp(S002): 8.68%

Insights:
With λ set to 0.5, the model reflects a moderate investor who equally values growth and stability. The optimizer allocates a majority of capital to Apple Inc. due to its superior return-to-risk ratio, while still diversifying slightly into Microsoft to reduce total portfolio volatility.
This modeling approach helps visualize the efficient frontier: by varying λ from 0 to 1, a series of optimal portfolios can be generated that map out the frontier of risk-return trade-offs. These

insights are invaluable for guiding clients toward portfolios that match their personal risk preferences.

Visualization of the Efficient Frontier:

To further illustrate the trade-off space between return and risk, we computed and plotted the efficient frontier by solving the multi-objective optimization problem under varying λ values from 0.0 to 1.0. Each point on the curve represents an optimized portfolio for a specific investor preference between risk and return.

Insights:

- As portfolio risk (standard deviation) increases, so does the expected return, following a convex shape.
- Lower-risk portfolios favor diversified, conservative assets, while higher-risk portfolios concentrate in high-return equities.
- This curve serves as a powerful decision tool: clients can select a preferred point along the frontier based on their risk tolerance.

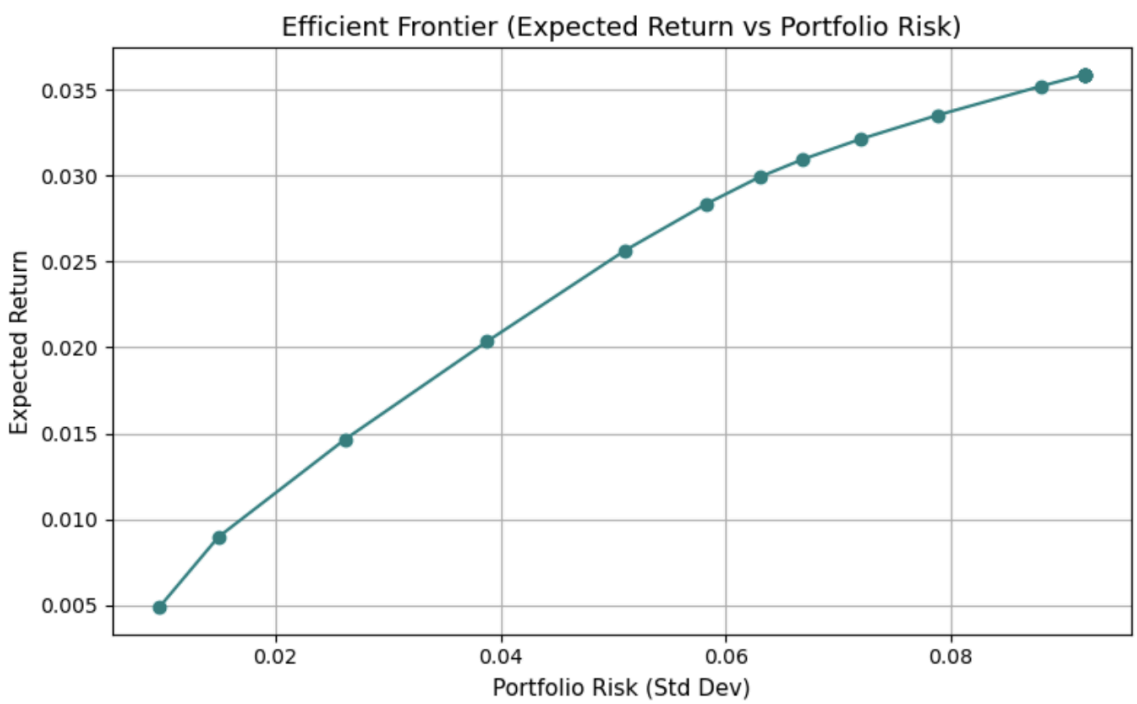


Figure: Efficient Frontier — Expected Return vs. Portfolio Risk

4. Strategic Insights

This study shows that effective portfolio optimization must be based on a deep understanding of client-specific characteristics, such as risk tolerance, investment horizon, and liquidity needs.

Our analysis confirms that certain assets, such as Apple, consistently dominate expected returns across models and scenarios.
However, there is a risk of excessive concentration in dominating the market, which could reduce the diversification goals of automated advisory systems and reduce portfolio diversity.
In addition, our results highlight the importance of constraint-sensitive optimization, especially for conservative investors who prioritize capital preservation and stability over aggressive growth. Therefore, portfolio design must consider both market behavior and investor-specific constraints to ensure that the solution is not only optimal in theory but also robust and effective in practice.

5. Final Recommendations

Based on these above key findings, we provide the following recommendations:

1. Use segment-specific optimization models

Applying different modeling approaches for different client types, like risk-based models, client-constrained models, or multi-objective models can help to ensure that each client receives a personal portfolio that matches their goals and limitations.

2. Implementing the client segmentation into onboarding

Using the clustering techniques during the onboarding process, which allows the systems to provide appropriate portfolio recommendations at the beginning.

3. Balance return with resilience

Combine performance-driven portfolio design with scenario testing and efficient frontier analysis which helps to ensure that portfolios remain stable across different economic conditions, not just in ideal scenarios.

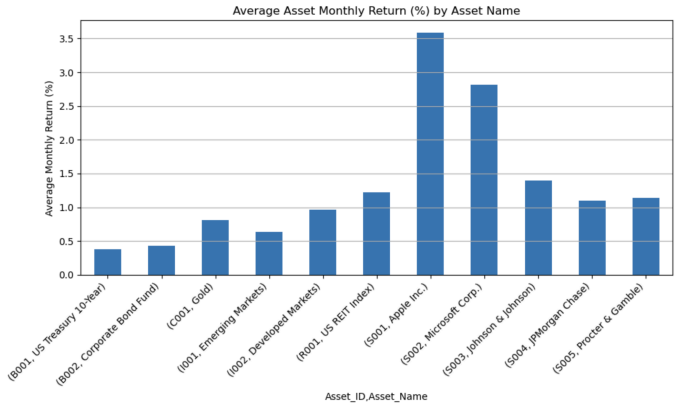
4. Enhancing the diversification and risk control constraints

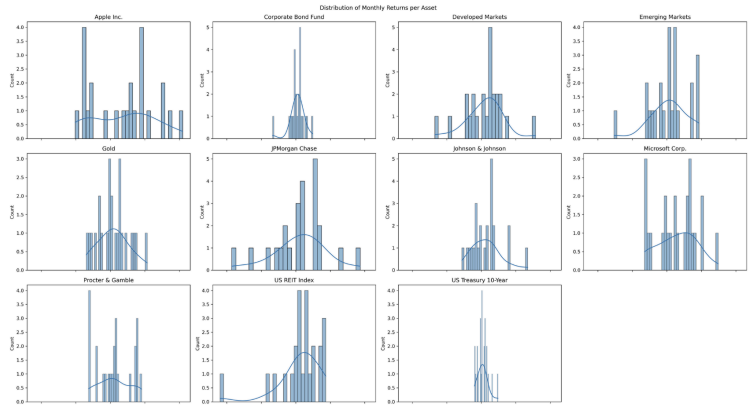
Applying and improving the appropriate constraint design in optimization models to avoid concentrate allocation to any single asset. This is especially important in automated portfolio systems to maintain balance and long-term sustainability.

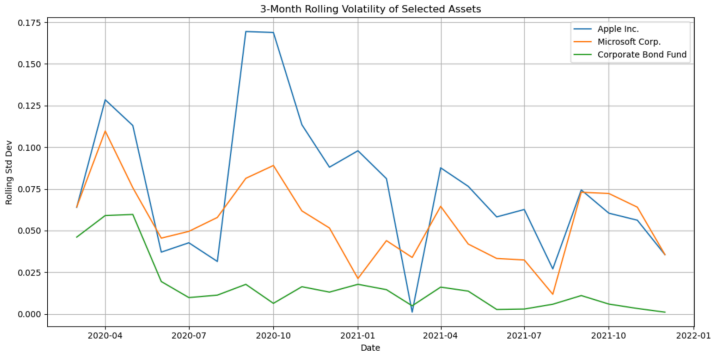
By appropriately adjusting through following these recommendations, financial advisors and systems can provide appropriate solutions that not only focus on optimizing the returns, but are also practical, diverse, and aligned with clients' specific requirements.

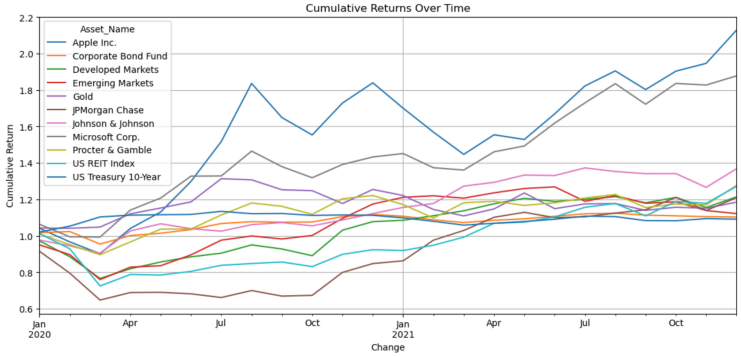
Appendix

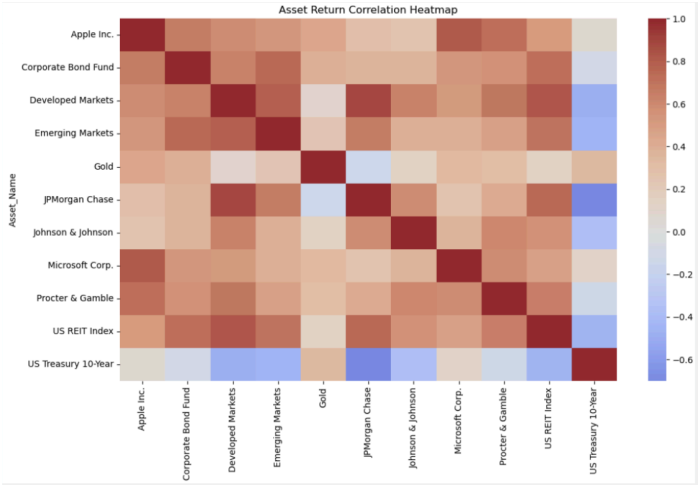
Asset-Level EDA Visualization:

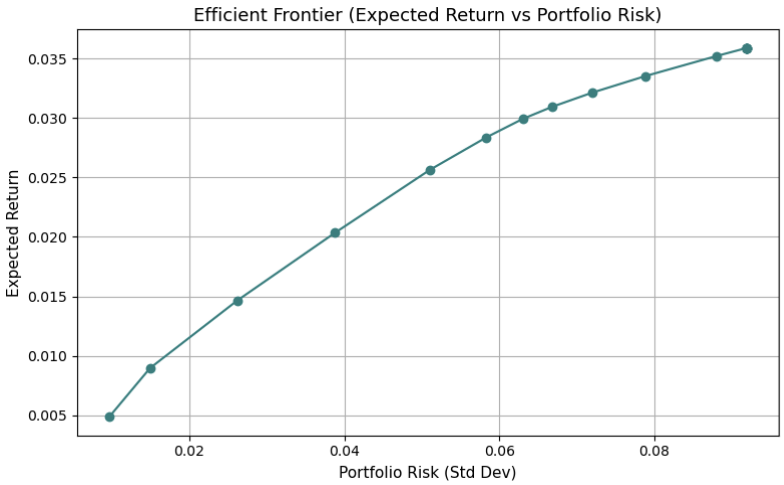
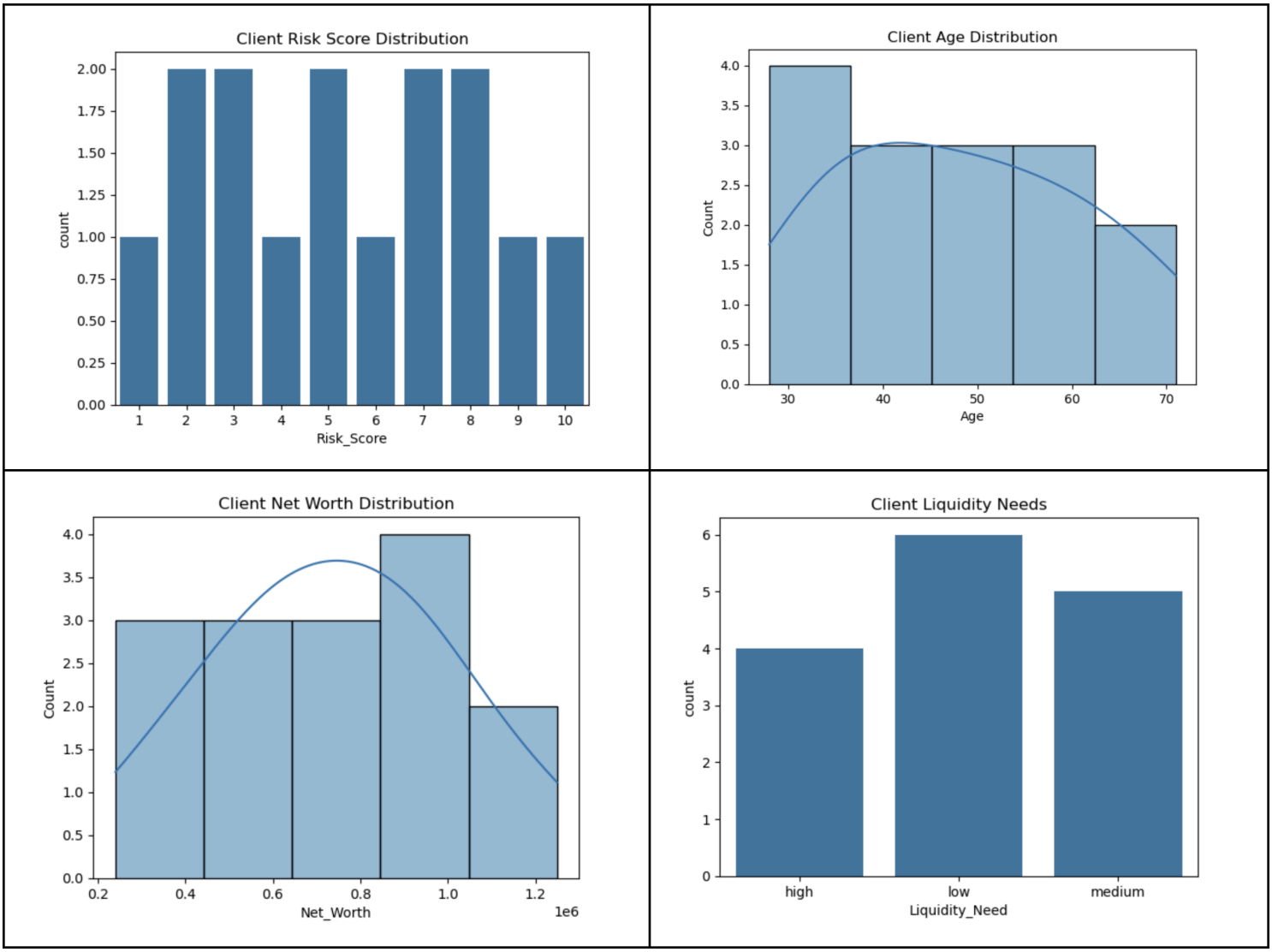












Client-Level EDA Visualization: